

Source Extraction Checkpoint S1: PaySim

This is the first application of the v1 source extraction template. PaySim was chosen as the entry point because it is the most accessible fraud-detection-oriented synthetic financial dataset and provides a baseline against which more authoritative sources (Tunisian regulatory and institutional data) will subsequently be calibrated.

Source Identification

- **Primary reference:** Lopez-Rojas, E. A., Elmir, A., Axelsson, S. (2016). *PaySim: A Financial Mobile Money Simulator for Fraud Detection*. In: 28th European Modeling and Simulation Symposium (EMSS), Larnaca, Cyprus.
- **Simulator codebase:** github.com/EdgarLopezPhD/PaySim (Java/MASON, GPL-3.0, 124 stars).
- **Public dataset (PaySim1):** Kaggle [ea1axi/paysim1](https://kaggle.com/ea1axi/paysim1), released 2017, ~6.36M records (1/4 scale of the paper's PS41840 dataset of ~23M records).
- **Operation types in source:** CASH-IN, CASH-OUT, TRANSFER, PAYMENT, DEBIT.
- **Temporal granularity:** step = 1 hour; coverage = 744 steps = 31 days.
- **Underlying real data:** aggregated logs from an unspecified African mobile money service. Currency anonymized.
- **Fraud injection:** not present in the original 2016 paper; added in subsequent Kaggle releases via `isFraud` and `isFlaggedFraud` columns.

Trust Profile (7 Dimensions)

Dimension	Score	Justification
Recency	MEDIUM	2016 paper, underlying data ~2013–2014, ~10–12 years old; transaction shape relatively stable over decade.
Authority	MEDIUM	Strong on academic methodology (Lopez-Rojas / Axelsson are well-cited), weak on regulatory authority (not a central bank or supervisory body).
Peer review	STRONG	EMSS 2016 (peer-reviewed conference, mid-tier simulation venue); hundreds of citations accumulated.
Sample size	MEDIUM	~6.36M synthetic records ample; size of underlying real aggregated logs not disclosed in paper.
Geographic relevance	WEAK	Unspecified African country; mobile money regulatory regime unrelated to Tunisian branch banking.
Data type	MEDIUM	Synthetic, calibrated from aggregated real parameters (two layers removed from ground truth).

Dimension	Score	Justification
Reproducibility	STRONG	Open-source simulator (GPL-3.0), public dataset, deterministic seed-based reproduction.

Tally: 2 Strong, 4 Medium, 1 Weak.

Verdict: Passes the trust gate as a *useful but heavily caveated* source. Parameters to be treated as starting points subject to cross-validation.

Operation Mapping

Our operation	PaySim type	Act match	Amount-shape match	Fraud-pattern match
Retrait	CASH-OUT	STRONG	MEDIUM	WEAK
Versement	CASH-IN	STRONG	MEDIUM	WEAK (no smurfing)
Virement	TRANSFER	STRONG	MEDIUM	WEAK
Chèque	—	NONE	NONE	NONE

Notes:

- PaySim's DEBIT type is parked as unused (mobile-to-bank flow; our virement schema does not distinguish internal/external).
- Chèque has no analogue in mobile money. This is the single largest gap in PaySim coverage.

Coverage Matrix (8 Dimensions)

Dimension	Confidence	Note
Amount distribution	MEDIUM	Shape transferable; absolute scale invalidated by anonymized currency.
Operation type mix	LOW	Mix measured precisely in PaySim, but mobile money mix does not transfer to branch banking.
Transaction frequency	LOW	Mobile-money frequency (multiple per day) is orders of magnitude above branch frequency (few per month).
Inter-event time	LOW	1-hour granularity floor; mobile timescale incompatible with branch timescale.
Temporal patterns	MEDIUM	Day-of-month payday spike transfers; hour-of-day distribution does not (mobile 24/7 vs branch hours).

Dimension	Confidence	Note
Spatial / branch	NOT COVERED	PaySim has no branch concept.
Channel mix	NOT COVERED	PaySim has no channel concept.
Beneficiary / emitter	HIGH	Graph structure (known-vs-new ratios, recurring destinations) is the most transferable signal PaySim offers.

Tally: 1 High, 2 Medium, 3 Low, 2 Not covered.

PaySim's practical role for our generator is as a **beneficiary-graph-structure source**, with secondary contributions on amount shape and day-of-month seasonality.

***Template gap noted:** The coverage matrix does not have a clean cell for operation-type sequence / transition patterns (Markov structure between successive operations). To be addressed in v2 of the template.*

Anomaly Typology Breakdown

PaySim fraud structure: A single injected pattern — account takeover via TRANSFER followed by CASH-OUT (cash drain). Concentrated in TRANSFER and CASH-OUT only. Overall fraud rate ~0.13% in the Kaggle dataset.

Bucket	Our typologies covered	PaySim signal
Direct	Virements détournés / retraits usurpés (account takeover via cash drain)	Pattern (TRANSFER → CASH-OUT sequence) + base rate (~0.13%) extractable.
Partial	Montant hors norme (virement); Bénéficiaire inconnu	Correlates with PaySim fraud label but not separately tagged.
Not covered	Smurfing; montants ronds suspects; cumul 48h; fréquence excessive; heure inhabituelle; retraits multiples GAB; virement en doublon; virement hors horaires; chèque sans provision; émetteur jamais vu; montant incohérent chèque; endossement à un tiers	Absent from PaySim.

Cross-cutting parameter: PaySim's overall anomaly base rate of ~0.13% is retained as a **starting calibration anchor** for the total anomaly fraction in our generated dataset, pending Tunisian-context validation. The base rate is logical in magnitude but not contextually equivalent.

Archetype Mapping

PaySim has no socioeconomic archetype labels in the public dataset (internal age segmentation — young / adult / senior — is not exposed). None of our five archetypes (salaried, small business, student, retiree, big business) receive direct calibration from PaySim.

Archetype	Coverage	Note
Salaried employee	NOT COVERED	No salary-deposit label in PaySim.
Small business owner	NOT COVERED	No business-vs-individual distinction.
Student	NOT COVERED	No demographic labels.
Retiree	NOT COVERED	No pension-deposit label.
Big business owner	NOT COVERED	High-volume clients exist but unlabeled.

Indirect contribution: PaySim's population-level per-client activity distribution is heavy-tailed, supporting a Layer-1 generator parameter for wide spread between low-activity and high-activity clients. Low confidence; does not assign behavior to specific archetypes.

Provenance Note

PaySim (Lopez-Rojas, Elmir, Axelsson — EMSS 2016) is an open-source agent-based mobile money simulator (MASON/Java, GPL-3.0) calibrated from aggregated transactional logs of an unspecified African mobile money service, with a public dataset of ~6.36M records released on Kaggle (CC-BY-SA). It is the **first extracted source** in a multi-source calibration strategy, not the most authoritative. PaySim's trust profile is mixed: strong on peer review and reproducibility, medium on recency, authority, sample size and data type, weak on geographic relevance (African mobile money, not Tunisian branch banking; anonymized currency precluding absolute-scale extraction). Its operation mapping covers three of our four operations weakly to medium (retrait ↔ CASH-OUT, versement ↔ CASH-IN, virement ↔ TRANSFER) and provides **no coverage of chèque operations**, our single largest gap. From the 8-dimension coverage matrix, PaySim is best understood as a **beneficiary-graph-structure source** (known-vs-new ratios, recurring-destination patterns transfer well), with secondary contributions to amount distribution shape (not scale) and day-of-month seasonality. It contributes nothing to spatial/branch patterns, channel mix, or archetype-specific parameters. On the anomaly side, PaySim covers a single typology directly — account takeover via TRANSFER → CASH-OUT cash drain, mapping to our virements détournés / retraits usurpés — and provides an overall anomaly base rate of ~0.13% retained as a starting calibration anchor pending Tunisian-context validation. All parameters extracted from PaySim are to be treated as starting points subject to cross-validation against at least one additional source before being locked into the generator.

Extracted Parameters Summary

Usable parameters (subject to cross-validation):

- Per-operation amount distribution *shape* (skew, coefficient of variation, percentiles) for retrait/versement/virement.
- Beneficiary graph structure: known-vs-new destination ratios, recurring-destination degree distribution per client.
- Day-of-month seasonality: payday-spike effect in the first half of each month.
- Account takeover signature: sequential TRANSFER → CASH-OUT pattern as a labeled anomaly archetype.
- Overall anomaly base rate starting anchor: ~0.13%.
- Population activity-spread shape: heavy-tailed (Layer-1 hint).

Explicitly not extracted from PaySim:

- Absolute monetary amounts (anonymized currency).
- Operation type mix (mobile context, not branch).
- Hour-of-day distribution (24/7 mobile vs branch hours).
- Channel mix, spatial/branch patterns (not in PaySim).
- Smurfing / structuring patterns (different threshold regime).
- All chèque-related parameters and anomalies.
- Archetype-specific parameters.

Open Items and Next Steps

- **Cross-validation required** for every parameter marked Medium or above in the coverage matrix.
- **Chèque gap** remains unfilled — must be addressed by subsequent source(s); candidates include BCT (Banque Centrale de Tunisie) annual statistics on chèques sans provision.
- **v2 template revision:** add a coverage-matrix cell for *operation-type sequence / transition patterns*.
- **Next source to extract:** Amen Bank 2023 Annual Report (institutional, Tunisian context, expected to outrank PaySim on geographic relevance and authority).